

The Causal Relationship Between Economic Activity and CO₂ Emissions: A Panel Data Analysis

To what extent does economic activity impact CO₂ emissions per capita, and how do industrial competitiveness, fossil fuel electricity reliance, and manufacturing dynamics influence this relationship?

Introduction:

Understanding the relationship between economic activity and CO₂ emissions is crucial for assessing the environmental impact of economic growth. As economies expand, emissions typically rise due to increased production and energy consumption. This study aims to explore the causal effect of GDP per capita on CO₂ emissions per capita, shedding light on how economic activity contributes to climate change. Given the complexity and intricate nature of economic activity and related socio-environmental externalities, introducing key confounders to capture exogenous effects is necessary. In this study, I account for the effects of i) Competitive Industrial Performance Index (CIP); ii) Fossil Fuel Electricity (FFelec), and iii) the Manufacturing Value Added Share (MVAsh). With the aim to measure the impacts of industrial competitiveness, energy consumption patterns, and industrial dynamics. Thus, refining our understanding of their impact on the GDP- CO₂ emissions relationship.

1) Data and Measurements:

The dataset used in this analysis spans from 1992 to 2021 and covers 137 countries across various regions. The data was collected from multiple reliable sources, ensuring consistency and accuracy. The variables included in the dataset are sourced from several reputable platforms:

- GDP (PPP\$y) and CO₂ Emissions: These variables were obtained from the World Bank (for GDP) and Our World in Data (for CO₂ emissions). Both provide annual country-level statistics adjusted for purchasing power parity (PPP) and emissions per capita, respectively.
- CIP Index (Competitive Industrial Performance Index) and MVA Share (Manufacturing Value Added Share): These were sourced from the UNIDO (United Nations Industrial Development Organization) databases, which track industrial performance and structure across countries.
- FF Electricity (Fossil Fuel Electricity Use): This variable measures the total amount of electricity generated from fossil fuels (in terawatt-hours, TWh). It was sourced from Our World in Data, which provides detailed data on energy consumption patterns globally.

The dataset for Model 1 includes 137 countries from 1992 to 2021, with complete data. Model 2 also includes 137 countries, however, several countries had missing data, particularly before 2000, so the analysis for this model spans from 2000 to 2021. While I considered imputing missing values through regressive linear extrapolation, I ultimately decided against it to avoid potential biases in causal inference. A threshold of 8% missing data was applied during the cleaning process, with any countries exceeding this threshold being excluded from the study.

Table 1 provides a summary of the variables included in the analysis along with descriptive statistics for each variable:

Table 1: Summary of Variables and Descriptive Statistics

Variable	Description	Mean	Std Dev	Min	Max
Model 1: CIP Confounding Variable					
Year	Year of observation	-	-	1992	2021
Country	Number of countries	-	-	-	137
Model Variables					
GDP (PPP _y)	Gross Domestic Product PPP _y	23768.99	24592.51	510.82	145591.02
CO ₂ Emissions	Total CO ₂ emissions	5.58	7.27	0.02	76.62
Log GDP (PPP)	Log-transformed GDP	9.51	1.16	6.24	11.89
Log CO ₂ Emissions	Log-transformed CO ₂ emissions	0.86	1.56	-3.83	4.34
CIP Index	Competitiveness index (0-1 scale)	0.074	0.104	0.000	0.585
Model 2: Multiple Confounders (MvaSh and FFelec)					
Year	Year of observation	-	-	2000	2021
Country	Number of countries	-	-	-	137
Model Variables					
GDP (PPP _y)	Gross Domestic Product PPP _y	25393.16	25453.85	711.98	145591.02
CO _{2y} Emissions	Total CO ₂ emissions	5.56	7.02	0.02	67.58
Log GDP (PPP)	Log-transformed GDP _y	9.60	1.14	6.57	11.89
Log CO _{2y} Emissions	Log-transformed CO _{2y} emissions	0.90	1.52	-3.83	4.21
CIP Index	Competitive Industrial Performance (Index)	0.075	0.102	0.000	0.579
MVA Share	Share of Manufacturing Value Added (Index)	0.127	0.059	0.004	0.402
FF Electricity	Fossil fuel electricity use	95.23	402.34	0.00	5678.03

Model Overview and Variable Descriptions

- GDP (PPP\$y): Economic output adjusted for purchasing power parity, enabling cross-country comparisons.
- CO₂ Emissions: Total CO₂ emissions per capita, indicating environmental impact.
- Log-transformed Variables: Log-transformed GDP and CO₂ emissions to stabilize variance for regression.
- CIP Index: Measures industrial competitiveness, controlling for industrial structure and export capacity.
- MVA Share: Proportion of GDP from manufacturing, reflecting industrial economy structure.
- FF Electricity: Fossil fuel electricity use (TWh), highlighting energy mix and CO₂ impact.

2) Empirical Model:

To analyze the causal relationship between economic activity and **CO₂ emissions per capita**, I have employed various econometric **regressions**. Each approach is designed to address different methodological concerns, including omitted variable bias, time dependency, and structural variations. Thus, resulting in 6 regressions in model 1, with CIP as the only confounder.

1. Log-Log OLS Regression (Baseline)

The first regression is a simple log-log OLS model, which captures the elasticity of CO₂ emissions with respect to GDP. By log-transforming both variables, the coefficients can be interpreted as elasticities, meaning a 1% increase in GDP leads to a $\beta\%$ change in CO₂ emissions.

$$\log(CO_2y)_{it} = \beta_0 + \beta_1 \log(GDP_{PPP})_{it} + \epsilon_{it}$$

This model provides a general benchmark but does not control for industrial competitiveness or energy-related confounders. The lack of time and country controls means this regression may suffer from omitted variable bias.

2. Cross-Sectional OLS Regression (2005 & 2021)

To assess whether the GDP-CO₂ relationship has evolved over time, we estimate two separate cross-sectional regressions for the years 2005 and 2021.

$$\log(CO_2y)_i = \beta_0 + \beta_1 \log(GDPy)_i + \beta_2 val_cip + \epsilon_i$$

These models introduce the Competitive Industrial Performance (CIP) Index, which accounts for differences in industrial competitiveness. Since each regression uses a single year of data, these models do not account for temporal trends but provide insight into structural changes in emissions over time.

3. First-Differences Regression (No Lags)

To control for unobserved heterogeneity across countries, we employ a First-Differences (FD) model, which estimates the effect of GDP changes on emissions changes within a country over time.

$$\Delta \log(CO_2y)_{it} = \beta_1 \Delta \log(GDPy)_{it} + \beta_2 \Delta CIP\ val_cip + \beta_3 Time\ Trend_t + \epsilon_{it}$$

By differencing the data, country-specific, time-invariant factors (e.g., geography, culture) are removed. This assumes past GDP and CO₂ values do not influence present ones, which might not always hold. The inclusion of a time trend captures global changes in emissions.

4. First-Differences Regression (2-Year Lags)

To account for delayed effects of GDP on emissions, we estimate a two-year lagged First-Differences model:

$$\Delta \log(CO2y)_{it} = \beta_1 \Delta \log(GDPy)_{it-2} + \beta_2 \Delta val_cip_{it-2} + \beta_3 Time\ Trend_t + \epsilon_{it}$$

This regression assumes economic changes do not immediately impact emissions but materialize two years later. A lagged structure helps capture short-term spillover effects. This approach ensures that the estimated effects of GDP changes on CO₂ emissions are not just contemporaneous but include potential delayed responses.

5. First-Differences Regression (6-Year Lags)

To further explore the **long-run relationship** between economic activity and emissions, we introduce a **six-year lag**:

$$\Delta \log(CO2y)_{it} = \beta_1 \Delta \log(GDPy)_{it-6} + \beta_2 \Delta val_cip_{it-6} + \beta_3 Time\ Trend_t + \epsilon_{it}$$

This model captures the long-term impacts of economic changes on emissions. By allowing a six-year adjustment period, it accounts for structural shifts in industrial production and energy consumption. Useful for understanding whether economic growth drives permanent changes in emissions over a longer horizon.

6. Fixed Effects Regression (Panel Data)

Finally, we estimate a Fixed Effects (FE) model, which controls for time-invariant country-specific heterogeneity:

$$\log(CO2y)_{it} = \beta_1 \log(GDPy)_{it} + \beta_2 val_cip_{it} + \mu_i + \lambda_t + \epsilon_{it}$$

Where:

- μ_i represents country-fixed effects (e.g., geography, industrial structure).
- λ_t represents year-fixed effects (e.g., global economic shocks, climate policies).

This is the most robust regression, as it eliminates both country-specific and time-specific biases, making it the best approach for causal inference.

In model 2, once the additional confounders MVash and FFelec were incorporated, the regressions 2, 4, and 6, were ran:

2. Cross-section OLS for the year 2005:

$$\log(CO2y)_i = \beta_0 + \beta_1 \log(GDPy)_i + \beta_2 val_cip_i + \beta_3 MVAsh_i + \beta_4 FFelec_i + \epsilon_i$$

4. First-Differences Regression (2-Year Lags):

$$\Delta \log(CO_2y)_{it} = \beta_1 \Delta \log(GDPy)_{it} - 2 + \beta_2 \Delta valcipit - 2 + \beta_3 TimeTrendt + \epsilon_{it}$$

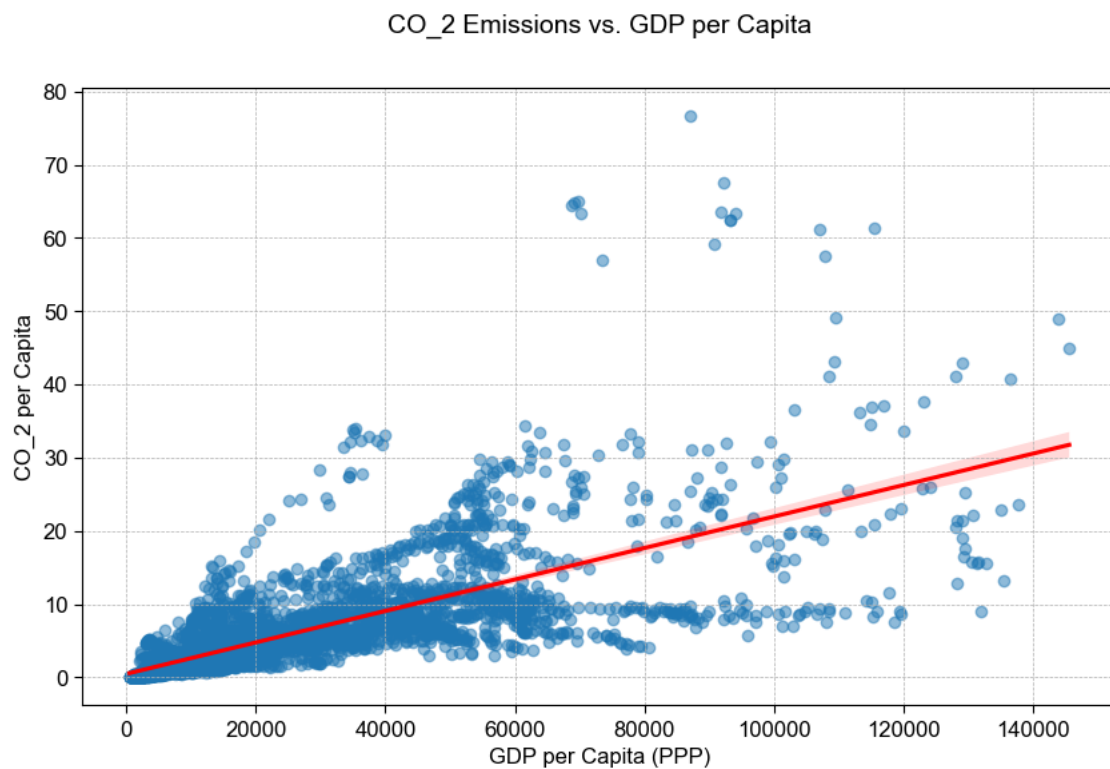
6. Fixed Effect for Year and Country with all Confounders

$$\log(CO_2y)_{it} = \beta_1 \log(GDPy)_{it} + \beta_2 valcipit + \beta_3 MVAshit + \beta_4 FFelecit + \mu_i + \lambda_t + \epsilon_{it}$$

3) Results and Discussion

a. Results

Graph 1. CO₂ Emissions vs. GDP per Capita (Scatter Plot with Regression Line)



This graph illustrates the positive relationship between GDP per capita and CO₂ emissions per capita across countries. Resulting from the Baseline logarithmic regression in Model 1.

Table 2 presents the regression results for CO₂ emissions per capita as a function of GDP per capita across various econometric specifications. The table begins with a log-log OLS regression, providing a baseline elasticity estimate of emissions concerning GDP. This is followed by two cross-sectional OLS regressions for 2005 and 2021, allowing us to assess whether the emissions-GDP relationship has changed over time. The next three models employ first-differences regressions, which eliminate time-invariant country-specific biases. These models estimate contemporaneous, two-year lagged, and six-year lagged effects of GDP on emissions, providing insight into both immediate and delayed relationships. Lastly, the fixed effects model accounts for both country- and year-fixed effects, offering the most robust causal estimate of the GDP-CO₂ relationship.

Table 2: Regression Results: CO₂*y* Emissions and GDP*y*

	Log-Log OLS	CS OLS (2005)	CS OLS (2021)	FD (No Lags)	FD (2-Yr Lags)	FD (6-Yr Lags)	Fixed Effects
Intercept	-10.8808 (0.095)	-11.5060 (0.483)	-10.1728 (0.518)	-0.1636 (0.405)	0.9464 (0.430)	0.8808 (0.517)	-5.0616 (0.183)
Log GDP	1.2389 (0.010)	1.3073 (0.053)	1.1487 (0.055)	0.6306 (0.031)	0.1349 (0.033)	0.1445 (0.032)	0.5539 (0.021)
CIP Index	-0.4626 (0.116)	-0.6013 (0.571)	-1.3951 (0.747)	0.5369 (0.279)	0.5130 (0.286)	-0.2523 (0.278)	1.7722 (0.183)
Time Trend	- -	- -	- -	0.00008 (0.000)	-0.0005 (0.000)	-0.0004 (0.000)	- -
Observations	4110	137	137	3973	3699	3151	4110
R ²	0.827	0.864	0.814	0.102	0.008	0.007	0.981

Table 3 compares the goodness-of-fit metrics across all baseline models. The R² and adjusted R² values indicate how well each specification explains the variation in CO₂ emissions. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) provide additional model comparison criteria, where lower values indicate better model fit. This table helps identify which model specification provides the best balance between explanatory power and parsimony.

Table 3: Comparison of Model Fit Metrics

	R ²	Adjusted R ²	AIC	BIC
Log-Log OLS	0.827	0.827	8095	8114
CS OLS (2005)	0.864	0.862	253.3	262.1
CS OLS (2021)	0.814	0.811	244.7	253.5
FD (No Lags)	0.102	0.102	-6642	-6617
FD (2-Yr Lags)	0.008	0.007	-6494	-6469
FD (6-Yr Lags)	0.007	0.006	-5926	-5902
Fixed Effects	0.981	0.980	-730.0	332.0

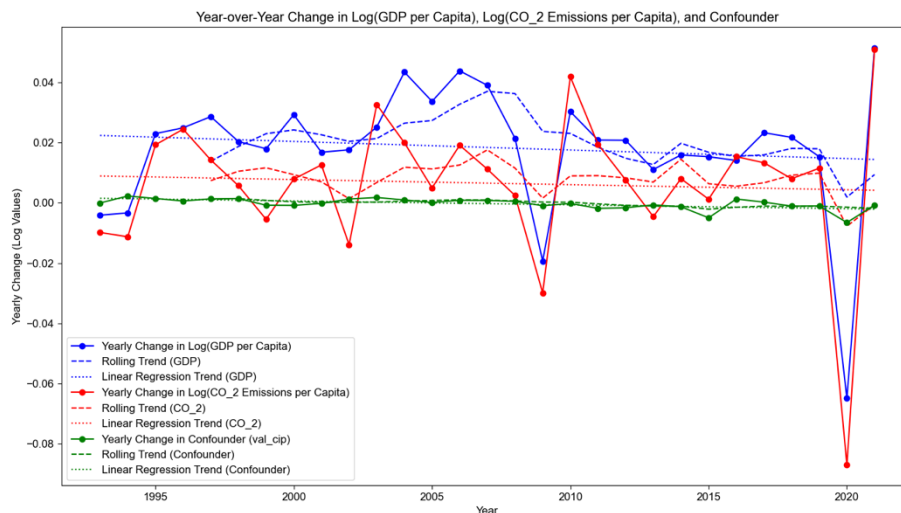
Table 4: Regression Results: CO_{2y} Emissions and GDP_y (With Additional Confounders)

	CS OLS (2005)	FD (2-Yr Lags)	Fixed Effects
Intercept	-12.2350 (0.511)	1.0237 (0.640)	-5.9114 (0.223)
Log GDP	1.3685 (0.053)	0.1070 (0.041)	0.6475 (0.025)
CIP Index	-2.2535 (0.673)	0.7857 (0.309)	3.1157 (0.224)
FFelec_val	0.0007 (0.000)	-9.328e-07 (4.78e-06)	-0.0001 (2.86e-05)
MVAsh	1.6806 (0.897)	-0.0155 (0.032)	-0.8890 (0.189)
Time Trend	- -	-0.0005 (0.000)	- -
Observations	137	2740	3014
R ²	0.880	0.008	0.988

Lastly, the results displayed in table 4, extend the analysis by incorporating two additional confounders: **Manufacturing Value Added Share (MVAsh)** and **Fossil Fuel Electricity Use (FFelec)**. These variables allow us to better isolate the independent effect of GDP on emissions while accounting for differences in energy production and industrial structure. The table reports results for three key models: **cross-sectional OLS for 2005**, **first-differences with a two-year lag**, and the **fixed effects model**. By including these confounders, Table 4 provides a more nuanced view of how industrial composition and energy mix shape the emissions-GDP relationship.

B) Discussions (Model Specific):

Graph2. Year-over-Year Changes (GDP, CO₂, CIP as confounder)



- I. Model 1 (Table 2): CIP as the only confounder:
- **Log-Log OLS Regression (Baseline Model):** The log-log OLS regression establishes the fundamental relationship between GDPs per capita and CO₂ emissions per capita. The elasticity estimates of 1.2389 ($p < 0.01$) suggests that a 1% increase in GDP per capita is associated with a 1.24% increase in CO₂ emissions per capita, confirming a strong and positive linkage between economic activity and environmental degradation. The high R² value (0.827) indicates that GDP alone explains a significant portion of CO₂ variation. Interestingly, the CIP coefficient is -0.4626 ($p < 0.01$), suggesting that higher industrial competitiveness is associated with lower emissions, potentially due to efficiency gains. However, as this model lacks controls for country- and time-specific factors, omitted variable bias is a concern.
 - **Cross-Sectional OLS (2005 & 2021):** In the cross-sectional regressions, the estimated elasticity is 1.3073 in 2005 and 1.1487 in 2021, indicating that the GDP-emissions relationship has slightly weakened over time. This could be attributed to increased adoption of cleaner technologies or structural economic shifts. The CIP coefficient is -1.3951 ($p > 0.1$) in 2021, suggesting a potential long-term shift where industrial competitiveness increasingly correlates with emissions reductions, though the statistical insignificance warrants caution in interpretation. The R² values for both years (0.864 and 0.814, respectively) indicate that GDP remains a strong predictor of CO₂ emissions.
 - **First-Differences (No Lags):** When first differences are applied to remove country-specific heterogeneity, the GDP elasticity shrinks to 0.6306 ($p < 0.01$), implying that economic growth has a less pronounced effect on emissions when considering within-country changes. The CIP coefficient flips to a positive value (0.5369, $p < 0.05$), suggesting that short-term increases in industrial competitiveness correlate with higher emissions. The time trend (0.00008, $p < 0.01$) is small but positive, indicating a slight upward drift in emissions. The R² value (0.102) shows that year-over-year changes in emissions are highly volatile, making GDP a weaker predictor in this model.
 - **First-Differences (2-Year Lag):** The GDP coefficient drops to 0.1349 ($p < 0.01$), indicating that GDP growth has a smaller delayed effect on emissions. The CIP coefficient remains positive (0.5130, $p < 0.05$), reinforcing the idea that industrial competitiveness drives emissions in the short term. The time trend (-0.0005, $p < 0.01$) suggests that global emissions growth is slightly slowing. However, the R² is very low (0.008), highlighting the difficulty of predicting emissions based on lagged GDP growth alone.
 - **Fixed Effects Model (Table 2):** The fixed effects model yields a GDP elasticity of 0.5539 ($p < 0.01$), lower than the OLS estimates, suggesting that once country-specific and time-fixed effects are accounted for, the effect of GDP on emissions is more modest. The CIP coefficient becomes strongly positive (1.7722, $p < 0.01$), implying that, when controlling for time-invariant factors, industrial competitiveness leads to increased emissions. This could indicate that more

competitive industries, despite efficiency gains, still rely heavily on carbon-intensive production. The R^2 (0.981) makes this the most robust model, explaining nearly all emissions variation.

II. Model 2. (Table 4) Multiple confounders:

Cross-Sectional OLS (2005) with Confounders: Incorporating manufacturing value-added (MVAsh) and fossil fuel electricity (FFelec) in the 2005 cross-sectional regression increases the GDP elasticity to 1.3685 ($p < 0.01$), suggesting that previous estimates may have been downward biased. The CIP coefficient turns strongly negative (-2.2535, $p < 0.01$), meaning that once energy mix and industrial structure are controlled for, industrial competitiveness is associated with reduced emissions. The FFelec coefficient (0.0007, $p < 0.01$) confirms that fossil fuel electricity increases emissions, as expected. The MVAsh coefficient (1.6806, $p > 0.1$) is positive but not significant, indicating that the manufacturing share does not independently drive emissions.

First-Differences (2-Year Lag) with Confounders: With additional confounders, the GDP coefficient remains small (0.1070, $p < 0.05$), reinforcing that economic growth has only a limited short-term impact on emissions when accounting for within-country variations. The CIP coefficient remains positive (0.7857, $p < 0.01$), implying that industrial competitiveness still increases emissions. Surprisingly, FFelec has an insignificant effect ($-9.328e-07$, $p > 0.1$), meaning that changes in fossil fuel electricity use do not significantly predict emissions at a two-year lag. The MVAsh coefficient is slightly negative (-0.0155, $p > 0.1$) but remains statistically insignificant.

Fixed Effects Model with Confounders: The final model, incorporating all confounders while controlling for country and time fixed effects, yields a GDP elasticity of 0.6475 ($p < 0.01$), slightly higher than in the baseline fixed effects model. The CIP coefficient remains strongly positive (3.1157, $p < 0.01$), reinforcing the finding that industrial competitiveness correlates with higher emissions once country-specific and temporal factors are considered. Interestingly, FFelec remains negligible (-0.0001 , $p > 0.1$), suggesting that fossil fuel electricity use is less relevant in explaining within-country emissions trends. However, MVAsh has a significant negative effect (-0.8890, $p < 0.01$), implying that an increased share of manufacturing in GDP reduces emissions, potentially due to improved efficiency or shifts toward less carbon-intensive industries. The R^2 (0.988) confirms this as the best-fitting model.

4) Conclusion

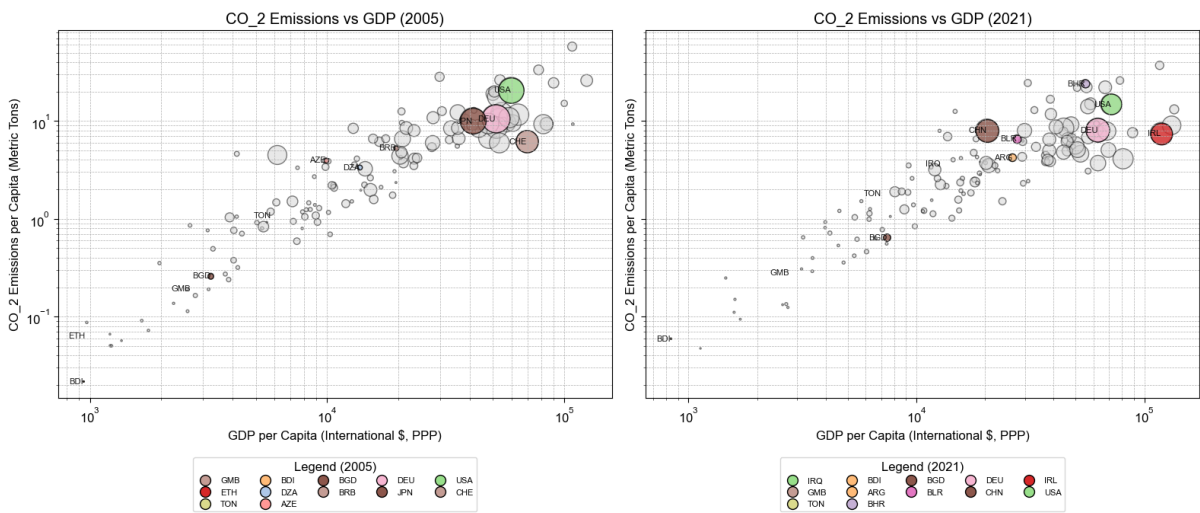
This study examines the causal relationship between economic activity (GDP per capita) and CO₂ emissions per capita over the period 1992–2021, using data from 137 countries. The core research question is: To what extent does economic activity cause CO₂ emissions? To address potential biases, key confounders were introduced: Competitive Industrial Performance (CIP) Index, Fossil Fuel Electricity Use (FFelec), and Manufacturing Value Added Share (MVAsh). These variables account for differences in industrial competitiveness, energy reliance, and structural composition of economies, ensuring a more refined estimation of GDP's impact on

emissions. The empirical strategy involved six model specifications: a log-log OLS regression, two cross-sectional OLS models (2005 & 2021), three first-differences regressions (capturing short-term and lagged effects), and a fixed effects panel model, which offers the most robust causal inference by controlling for unobserved heterogeneity.

Results indicate that GDP growth consistently increases CO₂ emissions, with an elasticity of 1.24 in the OLS model but declining to 0.55 in the fixed effects model, suggesting that once country-specific factors are controlled for, economic growth is less carbon-intensive. CIP's impact varies—negative in OLS (suggesting efficiency gains) but positive in panel models (implying emissions growth with industrial expansion). Fossil fuel electricity use is positively correlated with emissions, though its effect diminishes in fixed effects models, suggesting energy transitions. MVAsh reduces emissions in panel models, implying that shifting toward manufacturing could be a path to lower carbon intensity. Policy-wise, results highlight the importance of energy-efficient industrial policies and clean energy transitions. However, limitations include data gaps before 2000 and potential endogeneity concerns in industrial variables. Future research should consider instrumental variable approaches and sector-specific emissions trends to refine causal interpretations.

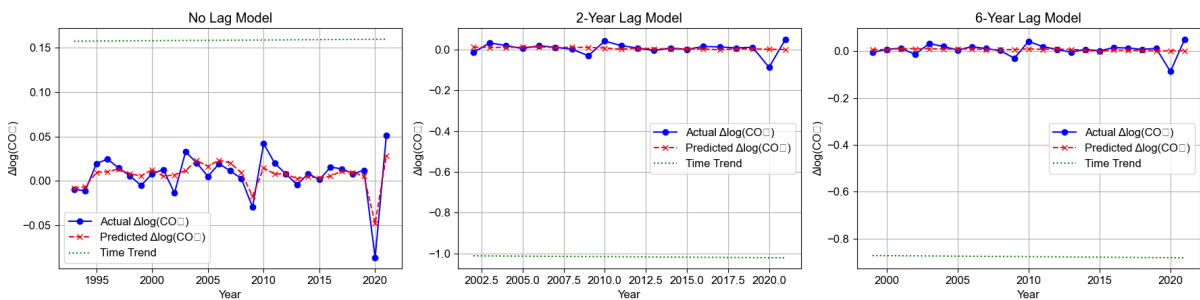
Annex:

Graph 3. CO₂ Emissions vs. GDP (2005 vs. 2021).



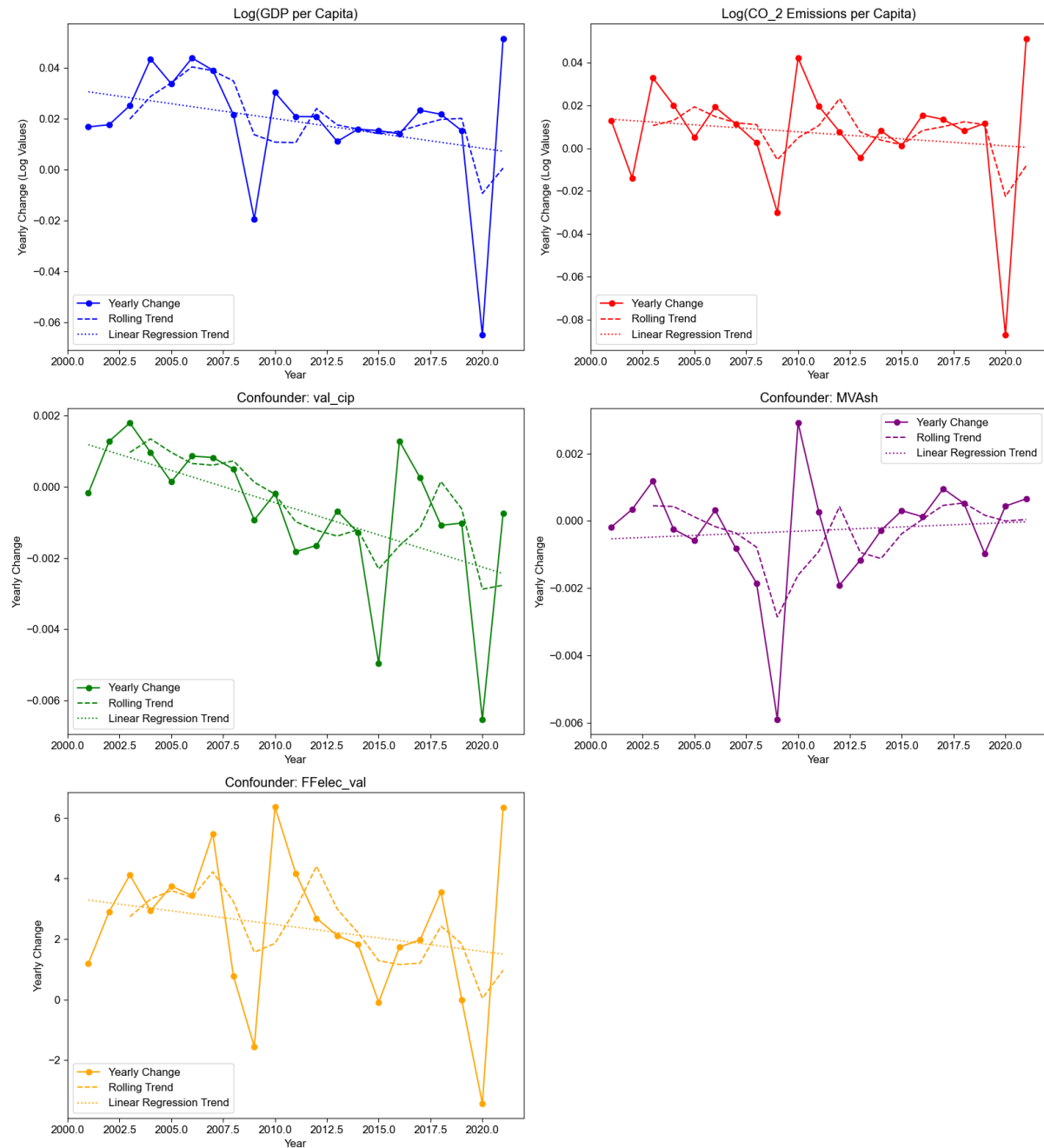
Graph 4. Trends in $\Delta \log(\text{CO}_2)$ (No Lag, 2-Year Lag, 6-Year Lag)

Overall Trends in $\Delta \log(\text{CO}_2)$ with Time Trend



Graph 5. Year-over-Year Changes in GDP, CO₂, and all confounders (model 2)

Year-over-Year Changes and Trends for GDP, CO₂ Emissions, and Confounders



Note: for additional graphs and if interested in per country x and y trend variations and fixed effect values, see Jupiter notebook.